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Course: DS Fundamentals with Python, Learners' Space 2026

Phase: Phase 1 — Python Modules

✓ Question 1

Create a 5x5 NumPy array with random integers between 1 and 100 and perform the following tasks:

- **(a)** Extract the middle element
- **(b)** Mean of each row
- **(c)** Elements greater than overall mean
- **(d)** Spiral order traversal function

```
import numpy as np

arr = np.random.randint(1, 101, size=(5, 5))
print(arr)
```

```
[[93 47 65 45 83]
 [86 19 18 63 43]
 [79 11 8 49 78]
 [43 23 66 42 4]
 [78 26 3 60 48]]
```

✓ (a) Middle Element

```
middle = arr[2, 2]
print("Middle element:", middle)
```

Middle element: 8

✓ (b) Mean of Each Row

```
row_means = np.mean(arr, axis=1)
for i, m in enumerate(row_means):
    print(f"Row {i}: {m}")
```

```
Row 0: 66.6
Row 1: 45.8
Row 2: 45.0
Row 3: 35.6
Row 4: 43.0
```

✓ (c) Elements Greater Than Overall Mean

```
overall_mean = np.mean(arr)
print("Overall mean:", overall_mean)
filtered = arr[arr > overall_mean]
print("Elements greater than mean:", filtered)
```

```
Overall mean: 47.2
Elements greater than mean: [93 65 83 86 63 79 49 78 66 78 60 48]
```

✓ (d) Spiral Order Function

```
def numpy_spiral_order(matrix):
    result = []
    top, bottom, left, right = 0, matrix.shape[0] - 1, 0, matrix.shape[1] - 1
    while top <= bottom and left <= right:
        for col in range(left, right + 1):
            result.append(matrix[top, col])
        top += 1
        for row in range(top, bottom + 1):
            result.append(matrix[row, right])
```

```

right -= 1
if top <= bottom:
    for col in range(right, left - 1, -1):
        result.append(matrix[bottom, col])
    bottom -= 1
if left <= right:
    for row in range(bottom, top - 1, -1):
        result.append(matrix[row, left])
    left += 1
return result

```

```

spiral = numpy_spiral_order(arr)
print("Spiral order:", spiral)

```

Spiral order: [np.int64(93), np.int64(47), np.int64(65), np.int64(45), np.int64(83), np.int64(43), np.int64(78), np.int64(4)

Question 2

Using the **Video Game Sales** dataset to answer the following questions.

```

import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('/videogamesales.csv')
print(df.shape)
print(df.columns.tolist())
df.head()

```

```

(16583, 9)
['Name', 'Platform', 'Year', 'Genre', 'Publisher', 'NA_Sales', 'EU_Sales', 'JP_Sales', 'Other_Sales']

```

	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales
0	Asteroids	2600	1980.0	Shooter	Atari	4.00	0.26	0.0	0.05
1	Missile Command	2600	1980.0	Shooter	Atari	2.56	0.17	0.0	0.03
2	Kaboom!	2600	1980.0	Misc	Activision	1.07	0.07	0.0	0.01
3	Defender	2600	1980.0	Misc	Atari	0.99	0.05	0.0	0.01
4	Boxing	2600	1980.0	Fighting	Activision	0.72	0.04	0.0	0.01

Next steps: [Generate code with df](#) [New interactive sheet](#)

(a) Global Sales Column — Sort by Global Sales

```

df['Global_Sales'] = df['NA_Sales'] + df['EU_Sales'] + df['JP_Sales'] + df['Other_Sales']
df_sorted = df.sort_values('Global_Sales', ascending=False).reset_index(drop=True)
print(df_sorted.head(10).to_string())

```

	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sal
0	Wii Sports	Wii	2006.0	Sports	Nintendo	41.49	29.02	3.77	8.46	82.74
1	Super Mario Bros.	NES	1985.0	Platform	Nintendo	29.08	3.58	6.81	0.77	40.24
2	Mario Kart Wii	Wii	2008.0	Racing	Nintendo	15.85	12.88	3.79	3.31	35.83
3	Wii Sports Resort	Wii	2009.0	Sports	Nintendo	15.75	11.01	3.28	2.96	33.00
4	Pokemon Red/Pokemon Blue	GB	1996.0	Role-Playing	Nintendo	11.27	8.89	10.22	1.00	31.38
5	Tetris	GB	1989.0	Puzzle	Nintendo	23.20	2.26	4.22	0.58	30.26
6	New Super Mario Bros.	DS	2006.0	Platform	Nintendo	11.38	9.23	6.50	2.90	30.01
7	Wii Play	Wii	2006.0	Misc	Nintendo	14.03	9.20	2.93	2.85	29.01
8	New Super Mario Bros. Wii	Wii	2009.0	Platform	Nintendo	14.59	7.06	4.70	2.26	28.61
9	Duck Hunt	NES	1984.0	Shooter	Nintendo	26.93	0.63	0.28	0.47	28.31

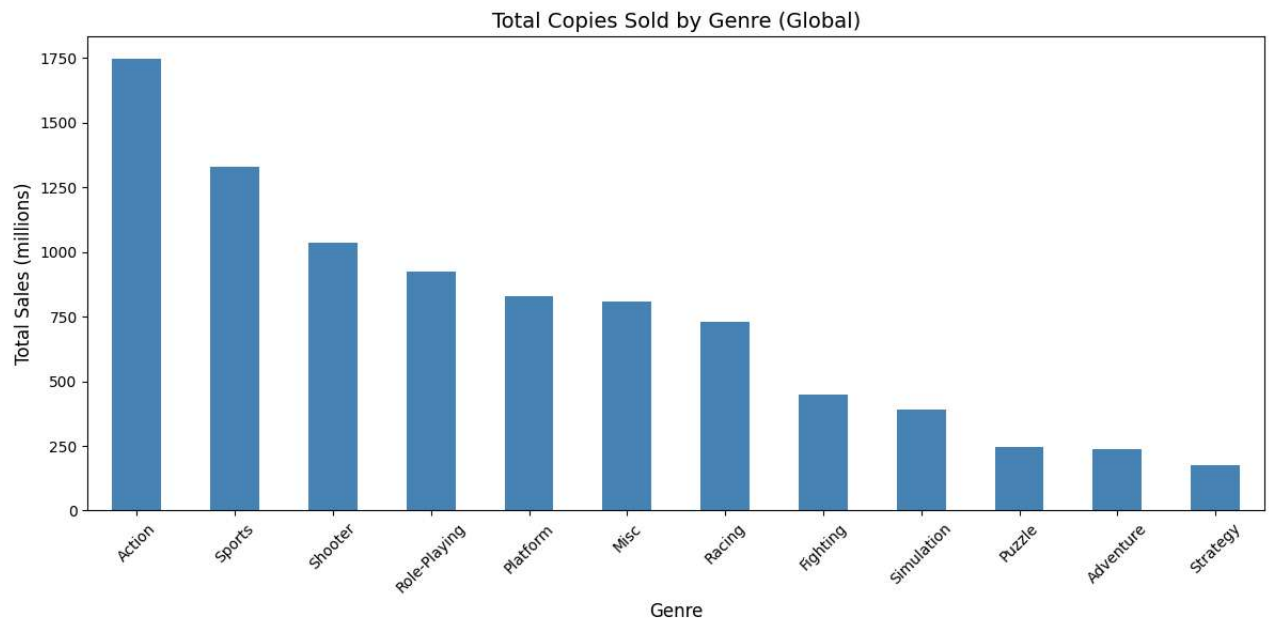
(b) Total Copies Sold by Genre (Global)

```

genre_sales = df.groupby('Genre')['Global_Sales'].sum().sort_values(ascending=False)

plt.figure(figsize=(12, 6))
genre_sales.plot(kind='bar', color='steelblue')
plt.title('Total Copies Sold by Genre (Global)', fontsize=14)
plt.xlabel('Genre', fontsize=12)
plt.ylabel('Total Sales (millions)', fontsize=12)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



✓ (c) Grand Theft Auto Games — Name, Platform, Year, EU+JP Sales

```
gta = df[df['Name'].str.contains('Grand Theft Auto', na=False)].copy()
gta['EU_JP_Sales'] = (gta['EU_Sales'] + gta['JP_Sales']).round(2)
gta_df = gta[['Name', 'Platform', 'Year', 'EU_JP_Sales']].reset_index(drop=True)
print(gta_df.to_string())
```

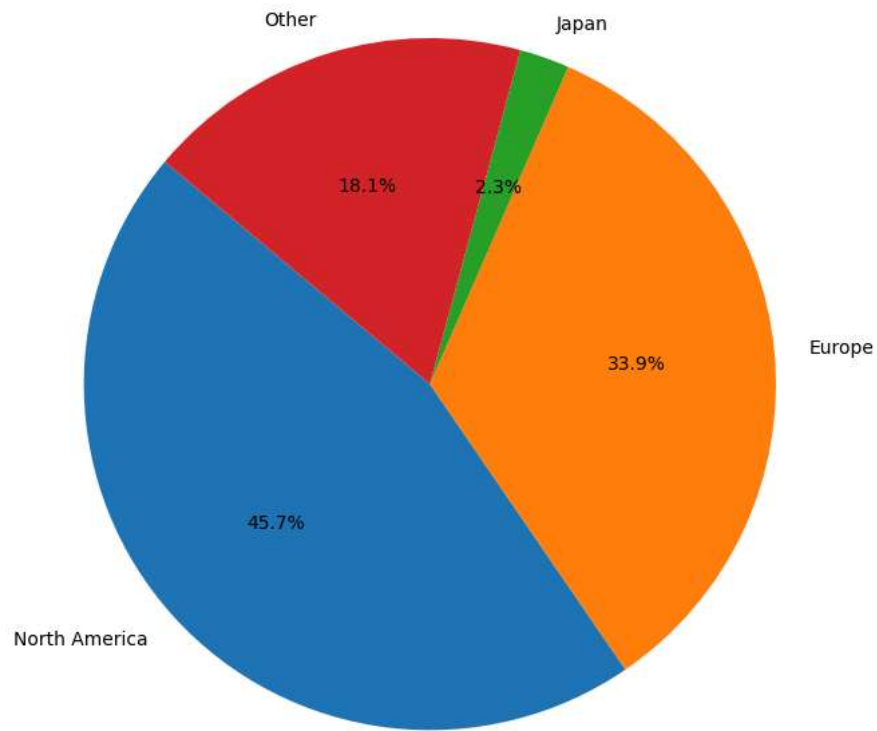
	Name	Platform	Year	EU_JP_Sales
0	Grand Theft Auto	PS	1997.0	1.39
1	Grand Theft Auto 2	PS	1998.0	2.07
2	Grand Theft Auto: Mission Pack #1, London 1969	PS	1998.0	0.04
3	Grand Theft Auto III	PS2	2001.0	4.81
4	Grand Theft Auto: Vice City	PS2	2002.0	5.96
5	Grand Theft Auto III	PC	2002.0	0.01
6	Rockstar Games Double Pack: Grand Theft Auto III & Grand Theft Auto Vice City	XB	2003.0	0.56
7	Rockstar Games Double Pack: Grand Theft Auto III & Grand Theft Auto Vice City	PS2	2003.0	0.66
8	Grand Theft Auto: Vice City	PC	2003.0	0.03
9	Grand Theft Auto: San Andreas	PS2	2004.0	0.81
10	Grand Theft Auto	GBA	2004.0	0.06
11	Grand Theft Auto: Liberty City Stories	PSP	2005.0	3.07
12	Grand Theft Auto: San Andreas	XB	2005.0	0.61
13	Grand Theft Auto: San Andreas	PC	2005.0	0.92
14	Grand Theft Auto: Vice City Stories	PSP	2006.0	2.18
15	Grand Theft Auto: Liberty City Stories	PS2	2006.0	1.47
16	Grand Theft Auto: Vice City Stories	PS2	2007.0	0.06
17	Grand Theft Auto IV	X360	2008.0	3.24
18	Grand Theft Auto IV	PS3	2008.0	4.20
19	Grand Theft Auto: San Andreas	X360	2008.0	0.03
20	Grand Theft Auto IV	PC	2008.0	0.79
21	Grand Theft Auto: Chinatown Wars	DS	2009.0	0.62
22	Grand Theft Auto: Chinatown Wars	PSP	2009.0	0.53
23	Grand Theft Auto V	X360	2013.0	5.37
24	Grand Theft Auto V	PS3	2013.0	10.24
25	Grand Theft Auto V	PS4	2014.0	6.17
26	Grand Theft Auto V	XOne	2014.0	2.01
27	Grand Theft Auto V	PC	2015.0	0.64

✓ (d) GTA Total Sales by Region — Pie Chart

```
gta_totals = gta[['NA_Sales', 'EU_Sales', 'JP_Sales', 'Other_Sales']].sum()
labels = ['North America', 'Europe', 'Japan', 'Other']
sizes = gta_totals.values

plt.figure(figsize=(7, 7))
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=140)
plt.title('Grand Theft Auto Total Sales by Region', fontsize=14)
plt.tight_layout()
plt.show()
```

Grand Theft Auto Total Sales by Region



Start coding or [generate](#) with AI.